



TY Coding

Lesson 02 – Variables and Datatypes

In the first lesson we learned

Characters: which are any single thing we can output using our keyboard.

Strings: which are groups of 2 or more characters.

Quotation marks: that characters and strings need to be surrounded by quotation marks.

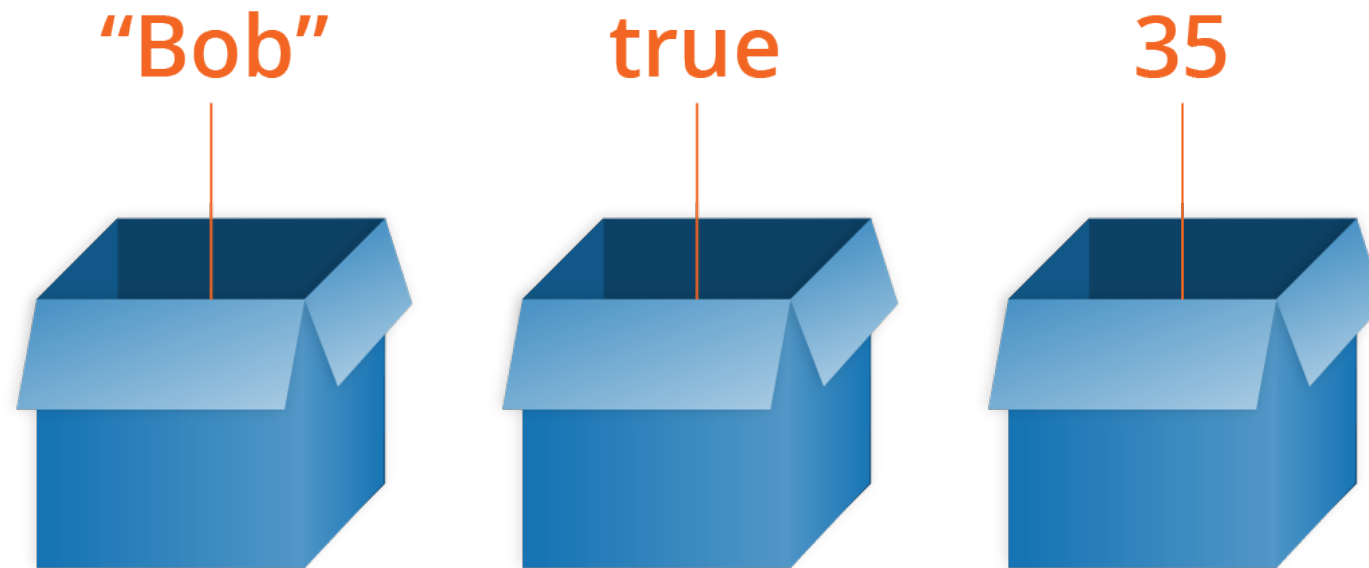
Functions: which are reusable pieces of code that do a task for us.

How to use the print function to output to our screens: print()

Lesson 01 Recap



Part 1 - Variables





What is a Variable

What is a variable?

- A variable is a container that we can create in our program to store a value.
- What values can a variable store?
- In python, there are different types of values which we call data types.
- A variable can be used to store any of these data types.

Data type	Short-hand name	Description of data type	Example of data type
Integer number	int	An int is a whole number	10
Floating point number	float	A float is a decimal number	3.14
Character	char	A single keyboard digit	"A"
String	str	A sentence or collection of characters	"My favourite food is pizza"
Boolean	bool	Holds the value of either True or False	True



Creating a variable

- To create a variable, we type (or "declare") the name of the variable, and then we assign a value to the variable.
- In python, the assignment operator is "=" which you know as the equals sign.
- Let's suppose we want to create one variable to store a string with our favourite animal in it and another variable to store our birth year.
- Which in our code editor, looks like:

```
My_favourite_animal = "hippopotamus"
```

```
My_birth_year = 2004
```

Variable name	Assignment operator	value
my_favourite_animal	=	"hippopotamus"
my_birth_year	=	2004



General RULES FOR NAMING YOUR VARIABLES

1. Can contain only numbers, letters and underscores.
2. Must not start with a number (but it may contain one).
 - a) My_income_1 is okay.
 - b) 1_my_income is not okay.
3. No spaces in the variable name.
4. No special characters (such as '@').
5. Must not be a python keyword (e.G. true, none, false, break).
6. Don't make your variable names too long.
7. The python naming convention is to use all lowercase letters, joined by underscores:
 - a) E.G. my_variable, my_age, house_width, number_of_pets



- EXAMPLES OF VARIABLES:
- `name = "john"`
`age = 15`
`latest_grade = "b"`
`grade_point_average = 3.8`
`is_a_student = true`
- The variable: `name` contains the string value: "john"
- The variable: `age` contains the integer value: 15
- The variable: `latest_grade` contains the character value: "b"
- The variable: `grade_point_average` contains the float value: 3.8
- The variable: `is_a_student` contains the boolean value: true
- *Nb* you will notice that you do not need "quotation marks" when assigning int, float & boolean values!



Variables and the print function

If you remember from last week, we used the `print()` function to print out strings. We can also use the `print()` function to print out variables! And the good news is that it works pretty much the same way.

Let's say we have 2 variables that contain our pets name and what year we are in, in school.

`pets_name = "bob ross"` remember we must use "quotation marks" for a string.

`pchool_year = 4` recall there are no need for quotation marks for an int, float or bool value.

So, let's print these variables using the print function:

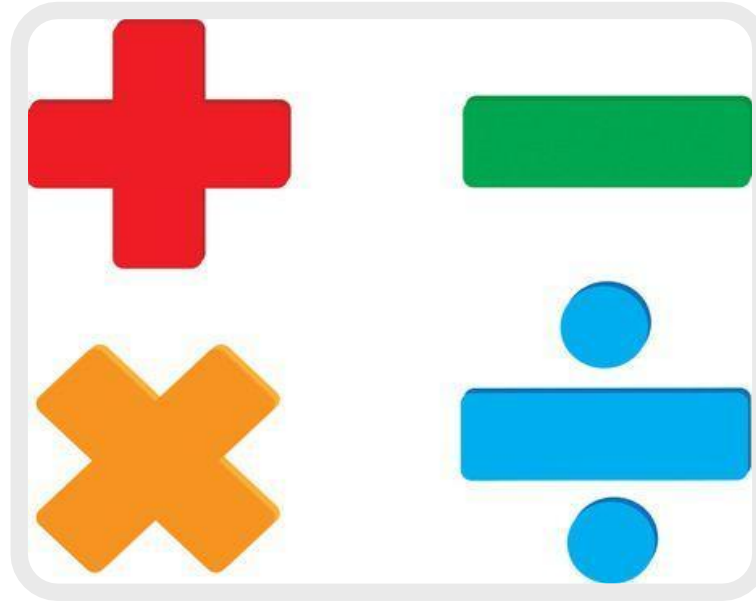
`print(pets_name)` we can simply place the variables

`print(school_year)` inside the print function to output them.

Will give the output:

Bob ross

4



Part 2 - Operators



- What are operators?
- There are different types of operators, we will be focusing on:
- Arithmetic operators
 - Arithmetic operators are used with numeric values to perform common mathematical operations. Addition, subtraction, multiplication etc...
- Assignment operators
 - Assignment operators are used to assign values to variables.
 - In python the "=" is the main assignment operator.



- Operators and the print function
 - We can use the `print()` function to print out the results of maths problems.
 - Let's say we wanted to know what $188 + 68 * 1 + 3$ was
 - We can use the python and the print function to tell us.
 - `Print(188 + 68 * 1 + 3)` will output the answer: 259
- And we can do this with:
 - Subtraction: `print(10-6)` will output the answer: 4
 - Multiplication: `print(5 * 4)` will output the answer: 20
 - Division: `print(100 / 20)` will output the answer: 5
- Python follows the pemdas order of operations. Which stands for:
 - Parentheses, exponent, multiplication, division, addition and subtraction.



Operator	Name	Description	Example	In words
+	Addition	Standard addition	$1 + 7 = 8$	One plus seven is eight
-	Subtraction	Standard subtraction	$5 - 3 = 2$	Five minus three is two
*	Multiplication	Standard multiplication	$6 * 4 = 24$	Six times four is twentyfour
/	Division	Standard division	$18 / 6 = 3$	Eighteen divided by six is three
%	Modulus	Like standard division, but the answer is the remainder.	$10 \% 3 = 1$	Ten divided by three, leaves a remainder of one
**	Exponentiation	Multiplies the left value to the power of the right value	$2 ** 5 = 32$	Two to the power of five is thirtytwo
//	Floor division	Like standard division, but the answer discards the remainder.	$10 // 3 = 3$	Ten floor divided by three is three and we discard the remainder of one.



Operator	Description	E.g. (number starts as 5)	Is the same as	Answers
=	Assigns a value to a variable	number = 5	The variable number contains a value of 5	number equals: 5
+=	Increment a variable by the amount specified after the = symbol	number += 7	number = number + 7	number now equals: 12 (number = 5 + 7)
-=	Deincrement a variable by the amount specified after the = symbol	number -= 3	number = number - 3	number now equals: 2 (number = 5 - 3)
*=	Multiply a variable by the amount specified after the = symbol	number *= 4	number = number * 4	number now equals: 20 (number = 5 * 4)
/=	Divide a variable by the amount specified after the = symbol	number /= 5	number = number / 5	number now equals: 1 (number = 5 / 5)
%=	Mod a variable by the amount specified after the = symbol	number %= 3	number = number % 3	number now equals: 2 (number = 5 % 3)
**=	Multiplies a variable to the power of the amount specified after the = symbol	number **= 4	number = 5 ** 4	number now equals: 625 (number = 5 ** 4)
//=	Floor divide a variable by the amount specified after the = symbol	number //= 2	number = number // 2	number now equals: 2 (number = 5 // 2)



Summary

- Today you learned about:
- Variables
 - Variables are containers that hold a value.
- Data types, that characters and strings are datatypes and some new datatypes:
 - Integers (whole numbers)
 - Floats (decimals)
 - Booleans (true or false)
- Operators
 - That you can use operators to assign values (e.G. The = assignment operator)
 - That you can use operators to do maths using python (e.G. The + addition operator)
- How to use the print function with variables and operators.